

Harsh Patnaik harshpatnaik133@gmail.com +91 8088271232 LinkedIn GitHub Location: Bengaluru, India Education PES University, Bengaluru B.Tech in Electronics and Communication Engineering Expected 2028	Technical Skills	
	Category	Technologies
	Robotics & Simulation	ROS 2, Gazebo, OpenCV, PID Control, Linux
	UAVs & Drone Tech	Betaflight, ELRS, Radiomaster Ranger Nano, ESC Configurator
	CAD & Engineering Design	SolidWorks (Surface Modeling), CFD Analysis, DfAM, Kinematics
	Embedded & IoT	ESP32, ESP8266, Arduino, Sensors (Gas, Ultrasonic), I2C/SPI Protocols
	Languages	Python, C, JavaScript, C++
	Web Development	React, Next.js, Node.js, Express.js, React Native, HTML, CSS
	Tools & Data	Git, MongoDB, Firebase, Pandas, Matplotlib, VS Code

Key Projects & Experience

Advanced STOVL/VTOL Endurance UAV | Active Research & Systems Architecture | Role: Lead Systems Designer | Tech: Powertrain Architecture, Aerodynamics, SolidWorks, CFD, Kinematics, Conceptual Design

- Aerodynamic & CFD:** Engineered a 2m BWB in SolidWorks; validated an L/D ratio of ~15 at 8° AoA.
- Powertrain & Propulsion Strategy:** Designed a thrust-vectoring system with phase-specific propellers, establishing kinematic torque balances and strict stator thermal caps for VTOL equilibrium.
- Performance Modelling & Material Specification:** Calculated conservative dynamic flight envelopes factoring in avionics overhead, and defined structural material specs to optimize rigidity and internal hardware volume.
- Commercial Viability & Modular Payload:** Developed a swappable "Smart Cartridge" belly interface for various payloads, maximizing enterprise utility.

e-Yantra Robotics Competition (eYRC 2025) – Stage 2 Qualifier | Role: Robotics & Drone Engineer | Tech: ROS 2, Python, Betaflight, Gazebo, ELRS Hardware

- Control Systems & Flight Stability:** Designed a PID UAV controller with velocity damping, thrust rate limits, and voltage feedforward to manage actuator nonlinearities and voltage sag.
- ROS 2 System Architecture:** Built a non-blocking ROS 2 Action Server/Client framework supporting concurrent navigation, real-time telemetry, and stabilization.
- Sim-to-Real Computer Vision:** Developed an OpenCV pipeline utilizing adaptive color thresholding for robust plant and asset detection under variable lighting.
- System Optimization:** Analyzed telemetry to tune filters and PWM ramping, mitigating oscillations and current spikes.
- Hardware Integration:** Resolved ESC mismatches, designed compact 555-timer latching circuits with LED feedback for reliable pesticide payload state retention.

Vega Racing Electric | Role: Electronics & Telemetry Engineer | Tech: Embedded Systems, Data Acquisition, LV Architecture, Wiring Harness, Telemetry

- Low-Voltage (LV) Architecture:** Collaborated on the low-voltage electronics hub (LV box) to serve as the centralized power distribution.
- Vehicle Control Unit (VCU):** Currently collaborating on real-time data acquisition and driver dashboard system designed to aggregate vehicle telemetry.

Recon Bot – Hackezee Hackathon 2025 (Runner-Up) | Role: Embedded Systems Lead | Tech: ESP8266, Sensors, IoT

- Prototyped a teleoperated hazard-zone robot in 8 hours featuring real-time sonar mapping, gas detection, and a cloud-based IoT sensor dashboard.

Terminal-Based Adventure Runner Game | Role: C Programmer | Tech: C Language, Data Structures

- Built a procedural terminal game using state machines, dynamic memory allocation, and function-pointer-based attack handlers, featuring scalable enemy generation, dodge probability logic, and persistent player progression via binary file storage.

Other Projects

- Voicelt – Grievance Portal (MERN Stack) - Top 10 Finalist, Codfusion Hackathon 2024:** Developed a full-stack grievance management platform with issue submission, categorization, tracking, and role-based admin tools.
- VigilAI – AI Misinformation Checker (Gemini API, Python, Web Scraping):** Built a multimodal prototype using Gemini API and Python to classify misinformation across YouTube/Instagram via text, audio, and keyframe extraction.

Research Interests

Autonomous Systems, UAV Dynamics & Control, Embedded Systems, EV architecture, IoT & Smart Devices, Robotics Simulation, Computer Vision.